

Sarthak Gupta

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EDUCATION

University of Florida

Bachelor of Science in Computer Science, Minor in Accounting (GPA: 3.68)

Gainesville, FL

Aug. 2024 – Dec. 2027

RELEVANT COURSEWORK

Introduction to Software Engineering
Advanced Programming Fundamentals
Introduction to Virtual Reality
Multivariable Calculus (III)
Information and Database Systems Design
Operating Systems

Computer Organization
Data Structures and Algorithms
Computational Linear Algebra
Discrete Structures
Engineering Statistics
Business Finance

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Java, Python, JavaScript, C++, R, MATLAB
Frameworks & Platforms: TypeScript, Angular.js, React.js, Node.js, Express.js, LangChain, FastAPI, TensorFlow, Robotics Operating System 2 (ROS2), PyTorch, React Native, Expo, Firebase, MongoDB, GIS
Tools & Technologies: Git, NVIDIA Isaac Sim, OpenCV, OpenMV, AWS Polly, Figma, Ubuntu/Linux, QGIS, ArcGIS Pro
Methodologies: SDLC, Agile, Scrum
Certifications: AWS Certified AI Practitioner, AWS Certified Cloud Practitioner, Building Transformer-Based NLP Applications (NVIDIA), Fundamentals of Deep Learning (NVIDIA), Intermediate Web Development (CodePath)

EXPERIENCE

Robotics Software Developer

Jan. 2025 – May 2026

Machine Intelligence Lab @ University of Florida

- Developed a C++ Gazebo plugin within the **ROS2 control framework** for the SubjuGator 9 submarine's gripper, integrating **JointController** and **JointTrajectoryController** for **2-DOF** velocity and position control; enabled full open-close actuation and trajectory-based manipulation in simulation.
- Integrated a Water-Linked Doppler Velocity Log (DVL) into the submarine using Bash, Linux, Python and C++; increased localization update rate to **10 Hz** and reduced drift from **12 m/hr** to **4 m/hr** during sea trials.
- Contributed to the Software Team in advancing SubjuGator 9 to the **semifinals of RoboSub 2025**, ranking **12th of 55** international teams by enhancing autonomy, sensor fusion, and reliability.

AI Developer Intern

Oct. 2025 – Present

Florida Resource Map Project @ Florida Community Innovation (FCI) Foundation

- Designing and deploying **LLM-driven agentic pipelines** using **LangChain**, **OpenAI APIs**, and **Python** to autonomously extract, validate, and structure non-profit and government service data from unstructured web sources.
- Building **web scraping and data-cleaning pipelines** to process over **5,000 resource listings** related to food, housing, and mental health; improving data coverage by **35%**.
- Implementing a **retrieval-augmented generation (RAG)** system with vector search for resource recommendations, increasing query accuracy and contextual relevance by **40%**.

COMPETITIONS

Vobile Innovation Challenge x IGNITE | Python, OpenCLAW, LangChain, Prompt Engineering

Mar. 2026 – Apr. 2026

- Placed **1st** out of all competing teams at the **CADE Museum Students Thinking Big Event**.
- Engineered a multi-stage **prompt chaining pipeline** using **OpenCLAW** and **LangChain** to autonomously generate, evaluate, and refine AI-driven creative concepts for Vobile's **DreamMaker** platform; implemented structured output schemas with **JSON mode** and **few-shot exemplars** to enforce consistent narrative generation across iterative refinement loops.
- Developed a **sliding-window visual grounding** algorithm that generated video in **5-second segments**, extracting a **reference keyframe** after each clip and re-injecting it as a **visual context anchor** into the subsequent prompt; eliminated cross-segment hallucination and maintained scene coherence across the full two-minute output.

PROJECTS

Epoch | React, Node.js, Express.js, Supabase (PostgreSQL), Google Gemini 2.5 Flash

2025 – Present

- Architected a full-stack AI history education platform across 15+ **Express.js** route modules with **role-based middleware** guards (teacher/student), **Supabase** auth, and **PostgreSQL jsonb** columns storing units, quizzes, assignments, notes, timelines, and AI-generated glossaries.
- Implemented an AI classroom assistant with **structured function-calling** (tool-use) for resource authoring and a multi-stage inference pipeline that grades free-text responses at 25-point resolution, then synthesizes cross-phase SA and MCQ results into scored wrapups with misconception detection and unit-level revisit recommendations.

JARVIS | Python, Gemini 2.5 Flash, openWakeWord, Whisper, Kokoro ONNX, PyQt6, Chrome MV3

2025 – Present

- Built a local, always-on voice assistant with a **multi-threaded daemon pipeline** — openWakeWord → Whisper STT → Gemini → Kokoro TTS — plus workers for **Gmail push watch**, scheduled notifications, screen watching, and a **Chrome MV3** bridge streaming live DOM/tab context into the model.
- Architected a **bounded tool-use orchestrator** over 30+ function-calling tools (shell, fs, git, **PyAutoGUI**, browser, Gmail) with malformed-call retry, **<ctrl*>** sentinel stripping, two-stage confirmation for destructive ops, and a direct-route fast path that bypasses the LLM.
- Designed a **two-tier memory** (short-term buffer + long-term JSON with tokenized top-K retrieval, importance decay, learned triggers) and a **vision loop** grounding **describe_screen/act_on_screen** on scoped screenshots fused with DOM snapshots.